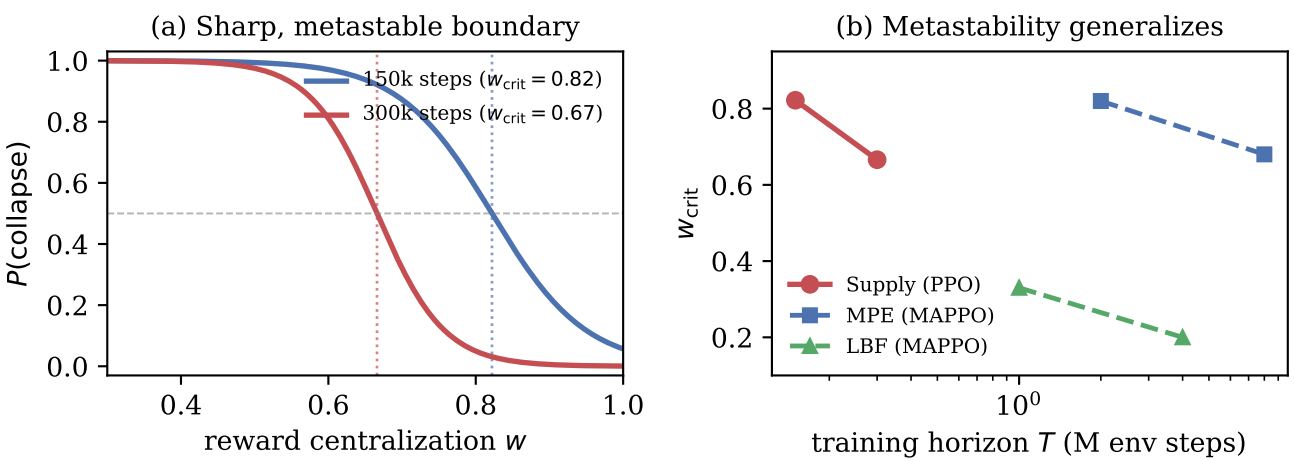
(a) Sharp, metastable boundary
1.0
0.8
0.6
0.4
0.2
0.0
P(collapse)
150k steps ($w_{crit} = 0.82$)
300k steps ($w_{crit} = 0.67$)
0.4
0.6
0.8
1.0
reward centralization w
(b) Metastability generalizes
0.8
0.6
0.4
0.2
w_{crit}
Supply (PPO)
MPE (MAPPO)
LBF (MAPPO)
10^0
training horizon T (M env steps)